Palatial Pools

FIELD TECHNICIAN MANUAL

Comprehensive Pool Services • Melbourne Pools, Made Easy

Pragmatic, Intelligent Pool Services — Smarter Solutions, Better Value

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HOW TO USE THIS MANUAL

This manual is designed like a field reference — fast to navigate under pressure. Use the section numbers and colour-coded headers to jump to the information you need. Purple headers mark major sections. Gold accents highlight key data and warnings. Decision trees give you step-by-step diagnostic paths. Quick Reference Cards at the back can be printed separately for your clipboard.

CONFIDENTIAL DOCUMENT

This manual is the property of Palatial Pool Services and is intended for use by authorised technicians only. Do not distribute, copy, or share externally. Contains proprietary procedures, dosing data, and operational standards.

01 Standard Service Procedure

Every Palatial Pools service visit follows a structured, repeatable procedure. Consistency is non-negotiable — it protects the client, protects you, and ensures nothing is missed. The procedure below applies to every regular service and single-visit service unless otherwise directed.

Service Visit Checklist (In Order)

- **1.** Arrive on site — visual inspection of pool condition, water clarity, debris load
- **2.** Photometric digital water test (before adding any chemicals)
- **3.** Empty skimmer basket(s) and deck canisters
- **4.** Empty pump basket (pump OFF first)
- **5.** Backwash filter (sand/glass) or remove and clean cartridge
- **6.** Vacuum pool floor, scoop surface debris, brush walls and steps
- **7.** Inspect equipment area — check for leaks, error codes, unusual noises
- **8.** Dose chemicals based on water test results (follow dosing order in Section 04)
- **9.** Adjust system parameters — chlorinator output, pump speed, timer settings
- **10.** Check and adjust water level (halfway up skimmer opening)
- **11.** Clean and replace robotic cleaner (if present)
- **12.** Final leak inspection around equipment pad and visible plumbing
- **13.** Record all findings, photos, and chemical additions in PoolTrackr

1.1 Arrival & Visual Assessment

Before touching anything, take 30 seconds to assess the pool visually. Note the water colour (clear, hazy, green, milky), the debris load on the surface and floor, and the water level. This initial assessment tells you what kind of service you are dealing with before the water test confirms it. If the pool is severely green or contaminated, adjust your approach accordingly — you may need to prioritise shock treatment over a standard clean.

> **PRO TIP — FIRST IMPRESSIONS MATTER**
>
> Take a photo of the pool on arrival. This documents the starting condition and protects you if a client later claims the pool was in better shape than it was. Upload all photos to PoolTrackr with the job notes.

1.2 Water Testing

Always test the water **before** adding any chemicals. Use the calibrated digital photometer — never dip strips. Dip strips rely on visual colour interpretation, which varies from person to person and is inherently unreliable. The digital photometer provides objective, numerical readings that can be tracked over time.

Weekly quick check: Free Chlorine and pH only. **Monthly full test:** All parameters including Total Alkalinity, Calcium Hardness, Cyanuric Acid, Phosphates, Salt, Combined Chlorine, LSI, and Temperature.

1.3 Cleaning Sequence

Always clean in this order: baskets first, then filter, then the pool itself. Cleaning baskets and the filter before vacuuming ensures the filtration system is operating at maximum efficiency when you need it most. Brush walls and steps before vacuuming the floor — brushing dislodges algae and debris that then settles to the floor where you can vacuum it up.

1.4 Chemical Dosing

Dose chemicals only after testing and cleaning. The correct dosing order is critical and is covered in detail in Section 04. Always allow the pump to circulate the water for at least 20 minutes between different chemical additions where possible. Record every chemical addition in PoolTrackr, including the exact product, quantity, and reason.

1.5 System Checks & Adjustments

Check the chlorinator is producing chlorine (verify cell is clean, check output percentage or ORP setting). Verify the pump timer is set to a single continuous block (see Section 08). Check for any error codes on controllers or automation systems. Ensure all valves are in the correct position for normal filtration.

1.6 Documentation

Every visit must be fully documented in PoolTrackr before leaving the site. This includes water test results, all chemical additions with quantities, service notes describing work performed, any issues identified, recommendations for the client, and photos of the pool condition and any notable findings.

> **NEVER LEAVE WITHOUT DOCUMENTATION**
>
> Incomplete records create liability. If it is not recorded, it did not happen. Every chemical addition, every observation, every recommendation must be in the system. This protects you, the client, and the business.

02 Water Testing & Interpretation

2.1 Testing Equipment

Palatial Pools exclusively uses calibrated digital photometers for water testing. These instruments measure chemical concentrations by passing light through a treated water sample and reading the absorbance. The result is a precise numerical value — not a colour match that depends on your eyesight, lighting conditions, or the age of the reagent.

> **CALIBRATION REMINDER**
>
> Calibrate the photometer at the start of every working day using the blank (untreated) water sample. If readings seem inconsistent, recalibrate immediately. Replace reagent tablets when they expire — expired reagents produce unreliable results.

2.2 Interpreting Results

Understanding what the numbers mean is just as important as getting accurate readings. The table below provides interpretation guidance for each parameter.

PARAMETER	LOW READING MEANS	HIGH READING MEANS	PRIORITY
Free Chlorine	Insufficient sanitisation; bacteria and algae can thrive	Potential skin/eye irritation; let it burn off naturally or dilute	Critical
pH	Corrosive water; damages equipment, etches surfaces, irritates skin	Chlorine becomes ineffective; scaling; cloudy water	Critical
Total Alkalinity	pH will bounce erratically; difficult to maintain stable chemistry	pH difficult to lower; cloudy water; scaling	High
Calcium Hardness	Water becomes aggressive; etches concrete/plaster surfaces	Scaling on surfaces and equipment; cloudy water	Medium
Cyanuric Acid	Chlorine burns off rapidly in sunlight; high chlorine consumption	Chlorine lock — chlorine present but ineffective	High
Salt	Chlorinator cannot produce chlorine; cell damage	Corrosion risk; salty taste; cell over-production	High (salt pools)
Phosphates	Good — no algae food source	Algae food source present; treat with phosphate remover	Medium
Combined Chlorine	Good — minimal chloramines	Chloramines present; causes smell and irritation; shock treat	Medium
LSI	Water is corrosive/aggressive to surfaces	Water is scale-forming	Monitoring

2.3 The pH-Chlorine Relationship

This is the single most important concept in pool chemistry. Free chlorine exists in water as two forms: **hypochlorous acid (HOCl)** — the active, effective sanitiser — and **hypochlorite ion (OCl^-)** — which is largely ineffective. The ratio between these two forms is entirely determined by pH.

PH LEVEL	% ACTIVE CHLORINE (HOCL)	% INACTIVE (OCL^-)	PRACTICAL EFFECT
7.0	73%	27%	Excellent kill rate but water is slightly acidic
7.2	63%	37%	Ideal — strong sanitisation, comfortable water
7.4	50%	50%	Good — acceptable balance
7.6	33%	67%	Chlorine efficiency dropping noticeably
7.8	22%	78%	Poor — significant sanitisation loss
8.0	**15%**	**85%**	**Chlorine is largely ineffective**
8.5+	**<8%**	**>92%**	**Chlorine almost completely ineffective**

CRITICAL RULE

At pH 8.0 or above, adding more chlorine is wasteful. You must lower the pH first with Hydrochloric Acid before the chlorine can do its job. High pH is one of the most common causes of algae and bacterial problems, even when chlorine levels appear adequate on the test.

2.4 Langelier Saturation Index (LSI)

The LSI is a calculated value that indicates whether the water is corrosive (negative LSI), balanced (near zero), or scale-forming (positive LSI). The target range is **-0.3 to +0.3**. The photometer calculates this automatically from pH, temperature, calcium hardness, total alkalinity, and TDS/salt. An LSI below -0.3 means the water is aggressively dissolving surfaces — particularly damaging to concrete and plaster pools. An LSI above +0.3 means calcium is precipitating out of solution and forming scale on surfaces and equipment.

03 Water Chemistry Targets

3.1 Palatial Pools Target Ranges

These are the target ranges for all Palatial Pools serviced pools. They are based on SPASA Victoria recommendations, AS3633, and our operational experience in Melbourne conditions.

PARAMETER	TARGET	ACCEPTABLE MIN	ACCEPTABLE MAX	UNIT
Free Chlorine	~4.0	2.0	10.0	ppm (mg/L)
pH	7.2 – 7.4	7.0	7.8	—
Total Alkalinity	80 – 120	80	250	ppm
Calcium Hardness (Concrete/Tile)	150 – 200	150	400	ppm
Calcium Hardness (Fibreglass/Vinyl)	125+	125	400	ppm
Cyanuric Acid (CYA)	30 – 50	30	50	ppm
Salt (Chlorinated Pools)	3000 – 5000	3000	5000	ppm
Combined Chlorine	<0.4	0	0.4	ppm
Phosphates	<0.4	0	0.4	ppm
LSI	0.0	-0.3	+0.3	—

3.2 Surface-Specific Considerations

Concrete / Quartzon / Tile Pools

Concrete-based surfaces are more tolerant of slightly aggressive water but are vulnerable to etching if the LSI drops too low. Maintain calcium hardness at **150–200 ppm minimum**. These pools tend to leach calcium into the water over time, which can push hardness levels up. Monitor and adjust accordingly.

Fibreglass / Vinyl Pools

These surfaces are more sensitive to chemical imbalance. Calcium hardness target is **125 ppm minimum**. Vinyl liners can be damaged by undissolved granular chlorine — always pre-dissolve granular chemicals in a bucket of pool water before adding. Fibreglass pools can develop osmotic blistering if water balance is consistently poor.

3.3 Cyanuric Acid — The Stabiliser Trap

Cyanuric acid (CYA) protects chlorine from UV degradation, which is essential for outdoor pools in Melbourne's climate. However, CYA accumulates over time because it does not break down or evaporate. Every time you add stabilised chlorine (dichlor or trichlor), you are also adding CYA.

> **CYA ABOVE 50 PPM**
>
> When CYA exceeds 50 ppm, it begins to bind with free chlorine and reduce its effectiveness — a condition known as **chlorine lock**. The only way to reduce CYA is to partially drain and refill the pool with fresh water. At CYA levels above 100 ppm (as seen in real Palatial Pools jobs), the chlorine is severely compromised regardless of the free chlorine reading.

04 Chemical Dosing Procedures

All chemicals used by Palatial Pools are sourced from **Pool Pro Australia** (www.poolpro.com.au). We use the Pool Pro range and Neptune range exclusively. Do not substitute generic or competitor products.

4.1 Dosing Order

When multiple parameters need correction, always dose in this order. This sequence is based on SPASA best practice and ensures each adjustment is effective.

1. **Total Alkalinity** — Adjust TA first. It acts as a buffer for pH. If TA is wrong, pH will bounce and resist correction. Use Pool Pro pH Buffer (sodium bicarbonate) to raise TA, or Hydrochloric Acid to lower it.
2. **pH** — Once TA is stable, correct pH. Use Pool Pro Hydrochloric Acid to lower pH, or Pool Pro pH Increaser (Soda Ash) to raise it. Wait 20 minutes with the pump running before re-testing.
3. **Sanitiser (Chlorine)** — Add chlorine only after pH is in range. Adding chlorine at high pH is wasteful because it converts to the inactive form. Use Pool Pro Stabilised Chlorine (granular), Liquid Chlorine, or Trichlor tablets depending on the system.
4. **Calcium Hardness & Cyanuric Acid** — Adjust these last. Use Pool Pro Hardness Raiser (calcium chloride) for low calcium. Use Pool Pro Rapid Stabiliser for low CYA. These are slow-acting and do not interfere with the above adjustments.
5. **Specialty Chemicals** — Phosphate remover, algaecide, clarifier, or flocculant as needed. These are added last and often require specific circulation times.

4.2 Dosing Reference Table

The following dosing rates are based on real Palatial Pools service data and Pool Pro product specifications. All doses are approximate and should be adjusted based on the specific pool volume and current readings.

PRODUCT	ACTIVE INGREDIENT	PURPOSE	DOSE PER 10,000L	APPLICATION METHOD
Pool Pro HCl Acid	Hydrochloric Acid ~32%	Lower pH & TA	300–350 mL	Pour slowly into return jet area with pump running
Pool Pro pH Buffer	Sodium Bicarbonate	Raise TA	750g – 1kg	Pre-dissolve in bucket, pour around pool edge
Pool Pro pH Increaser	Soda Ash (Sodium Carbonate)	Raise pH	100–200g	Pre-dissolve in bucket, pour around pool edge
Pool Pro Stabilised Chlorine	Sodium Dichlor / Trichlor	Raise Free Cl	125g (shock)	Pre-dissolve in bucket, pour around pool edge
Pool Pro Liquid Chlorine	Sodium Hypochlorite 12.5%	Raise Free Cl	500 mL	Pour directly into pool with pump running
Pool Pro Rapid Stabiliser	Cyanuric Acid	Raise CYA	50–250g	Pre-dissolve in warm water, add to skimmer
Pool Pro Hardness Raiser	Calcium Chloride	Raise Calcium	100–150g	Pre-dissolve in bucket, pour around pool edge
Pool Pro Phosphate Remover	Lanthanum Chloride	Remove Phosphates	200 mL	Pour near return jet with pump running
Pool Pro Algaecide	Quaternary Ammonium	Prevent/treat algae	Per label	Pour around pool perimeter
Pool Pro Clarifier	Cationic Polymer	Clear cloudy water	Per label	Pour near return jet

REAL-WORLD DOSING EXAMPLES FROM PALATIAL POOLS JOBS

40,000L Quartzon pool, pH 8.4: Added 1.3L Pool Pro HCl Acid.
40,000L Tile pool, Free Cl 0.07 ppm: Added 500g Pool Pro Stabilised Chlorine + 1.2L HCl Acid (pH was 8.3).
10,000L Fibreglass pool, TA 72 ppm: Added 3kg Pool Pro pH Buffer.
40,000L Tile pool, Phosphates 1.26 ppm: Added 800mL Pool Pro Phosphate Remover.

4.3 The Tropix Range

Pool Pro's **Tropix range** is a premium chemical line designed for specific applications. It includes enhanced formulations for clarifiers, algaecides, and water enhancers. Use Tropix products when standard Pool Pro products are insufficient or when a premium result is required (e.g., persistent algae problems, high-end residential pools requiring crystal clarity). Always check the Tropix product label for specific dosing instructions as they differ from the standard range.

05 Skimmer & Pump Basket Maintenance

5.1 Skimmer Baskets

The skimmer basket is the first line of defence in the filtration system. It catches large debris before it reaches the pump and filter. A full or clogged skimmer basket restricts water flow, reduces pump efficiency, and can cause the pump to lose prime.

Procedure

Remove the skimmer lid carefully — some lids are heavy or may be stuck. Lift the basket out and empty all debris into your waste bag. Inspect the basket for cracks or damage. A cracked basket allows debris to pass through to the pump, which can cause impeller blockages. Replace damaged baskets immediately. Check the weir door (flap) is moving freely — a stuck weir door reduces skimming efficiency. Ensure the skimmer throat is clear of any obstructions.

> **DECK CANISTERS**
>
> Some pools have deck-level canisters or leaf traps in addition to the standard skimmer. Always check and empty these as well. They are easy to miss but can cause significant flow restriction if full.

5.2 Pump Baskets

> **ALWAYS TURN THE PUMP OFF BEFORE OPENING**
>
> Never attempt to open the pump lid while the pump is running. The system is under pressure and the lid can blow off, causing injury. Turn the pump off at the controller or isolator switch, wait for the pressure to drop, then open the lid.

Procedure

Turn the pump off. Release the lid — some pumps have a twist-lock lid, others have a clamp. Remove the basket and empty debris. Inspect the basket for cracks. Check the lid o-ring for damage, cracks, or flattening. Apply silicone lubricant to the o-ring if it appears dry — this ensures a proper seal and prevents air ingress. Refill the pump basket housing with water before replacing the lid to assist with priming. Ensure the lid is seated correctly and tightened firmly but not over-tightened.

> **REAL-WORLD EXAMPLE — INV-1302**
>
> A tenant at a Balwyn property attempted to clean the pump basket themselves. They did not refill the housing with water and did not seat the lid correctly. This caused air ingress through the cleaner hose, and the pump seized. The technician had to re-prime the system, educate the tenant on the correct procedure, and check the o-ring for damage. Always educate clients on this process if they express interest in doing it themselves.

06 Filter Types & Backwashing

6.1 Sand / Glass Media Filters

Sand and glass media filters work by trapping particles as water passes through the filter bed. Over time, the trapped debris increases the pressure inside the filter. When the pressure gauge reads **50–70 kPa above the clean starting pressure**, or as part of a regular maintenance schedule, the filter needs to be backwashed.

Backwash Procedure

1. **Turn pump OFF.**
2. **Set multiport valve to BACKWASH.** Never move the multiport valve while the pump is running — this can damage the valve internals and the spider gasket.
3. **Turn pump ON.** Run for 2–3 minutes or until the sight glass runs clear.
4. **Turn pump OFF.** Set valve to RINSE.
5. **Turn pump ON.** Run for 30–60 seconds. This re-settles the media bed and flushes any remaining debris to waste.
6. **Turn pump OFF.** Set valve back to FILTER. Turn pump ON. Check pressure gauge — it should be back near the clean starting pressure.

> **BACKWASH DRAINAGE ISSUES**
>
> Some properties have drainage problems that prevent normal backwashing. At one Balwyn property (INV-1302), backwashing caused water to flood beneath the stairs. In these cases, all debris must be removed manually or by vacuuming to waste through an alternative drain point. Document the issue and advise the client that the backwash drainage setup must be repaired for proper ongoing maintenance.

6.2 Cartridge Filters

Cartridge filters use a pleated fabric element to trap particles. They do not require backwashing but must be physically removed and cleaned.

Cleaning Procedure

Turn the pump off. Release the filter housing clamp or lid. Carefully remove the cartridge element. Hose it down thoroughly with a garden hose, working from top to bottom, flushing debris from between the pleats. For a deeper clean (recommended every 4–6 weeks), soak the cartridge in a dedicated cartridge cleaner solution overnight to dissolve oils, sunscreen residue, and scale deposits.

REAL-WORLD EXAMPLE — INV-1301

At a 10,000L fibreglass pool in Oakleigh South, the cartridge filter was removed and cleaned during a regular service. After cleaning, the flow rate was measured at approximately 205 L/min at standard pump speed and 225 L/min at 120% pump speed. A flow meter was left in place for continued monitoring to ensure the cartridge maintained adequate performance between services.

6.3 Filter Comparison

FEATURE	SAND / GLASS MEDIA	CARTRIDGE
Filtration Fineness	20–40 microns (sand), 10–15 microns (glass)	10–15 microns
Maintenance	Backwash every 2–4 weeks	Remove and hose every 2–4 weeks
Media Replacement	Every 5–7 years (sand), 7–10 years (glass)	Every 1–3 years depending on use
Water Loss	Loses water during backwash (~200–400L)	No water loss during cleaning
Drainage Required	Yes — backwash line to drain	No
Best For	Larger pools, heavy debris loads	Smaller pools, low-debris environments

07 Equipment Setups & Configurations

7.1 Sanitisation Systems

Saltwater Chlorinator

The most common system in Melbourne residential pools. A salt chlorinator converts dissolved salt (NaCl) in the pool water into chlorine through electrolysis. The salt cell generates hypochlorous acid directly, which is the active sanitiser.

ASPECT	SALTWATER CHLORINATOR	LIQUID CHLORINE (MANUAL DOSING)
How It Works	Electrolysis converts salt to chlorine automatically	Sodium hypochlorite added manually to pool
Salt Required	3,000–5,000 ppm (check manufacturer spec)	Not required (some residual salt may be present)
Ongoing Cost	Low (salt is cheap; cell replacement every 3–7 years)	Higher (regular liquid chlorine purchases)
pH Tendency	Tends to push pH UP over time (requires regular acid dosing)	Tends to push pH UP (sodium hypochlorite is high pH)
CYA Impact	Does not add CYA (must add stabiliser separately)	Does not add CYA
Service Actions	Check cell condition, output %, ORP, salt level	Check chlorine drum level, dosing rate, top up as needed
Common Issues	Scaled cell, low salt, clock reset, cell failure	Empty drum, inconsistent dosing, no automation

CHLORINATOR CLOCK RESET

If a chlorinator's internal clock has reset (showing incorrect time/date), it usually means the unit lost power for an extended period and the internal battery has gone flat. This is common when properties are vacant between tenants. Reset the clock, check the output settings, and verify the cell is producing chlorine.

Trichlor Tablet Feeders

Some pools use an inline or offline tablet feeder that slowly dissolves trichlor tablets to maintain chlorine levels. These systems add CYA with every tablet, so monitor CYA levels closely. Tablet feeders are a set-and-forget solution but require regular tablet top-ups and CYA monitoring.

7.2 Heating Systems

FEATURE	HEAT PUMP	GAS HEATER
How It Works	Extracts heat from ambient air; transfers to pool water	Burns natural gas or LPG to heat a heat exchanger
Efficiency	Very high (COP 4–6); best in mild/warm weather	Lower efficiency but heats faster; works in any weather
Heating Speed	Slow — raises temp 1–2°C per day typically	Fast — can raise temp 5–10°C in hours
Running Cost	Low (electricity only)	Higher (gas consumption)
Common Issues	Thermal cycling, low flow cutout, ambient temp too low	Ignition failure, gas valve issues, error codes
Service Actions	Check flow rate, inlet valve position, mode settings, ambient temp	Check gas supply, error codes, ventilation. Certified gas work only.

REAL-WORLD EXAMPLE — HEAT PUMP THERMAL CYCLING (INV-1301)

A heat pump at a 10,000L fibreglass pool in Oakleigh South was set to 34°C but kept cutting out at approximately 29°C. The technician switched the unit from direct mode to automatic mode, increased the set temperature to monitor performance, and opened the inlet valve further to improve water flow. For the remainder of the visit, the unit operated without cutting out. If the issue recurs, it may indicate thermal cycling, requiring the unit to be relocated or further diagnosed.

7.3 Automation & Controllers

Many modern pools use automation systems (e.g., Astral Connect, Zodiac iQ, Pentair IntelliConnect) that control the pump, chlorinator, heater, and lighting from a single controller or smartphone app. When servicing these systems, check for error codes, verify scheduled run times, and ensure the chlorinator output is set appropriately. Do not change automation settings without understanding the full system configuration and documenting the changes.

7.4 Robotic Cleaners

Many Palatial Pools clients have robotic pool cleaners. These are independent units that operate on their own power supply and do not connect to the pool's filtration system. During each service visit, remove the robot from the pool, empty its filter bag or cartridge, rinse it with a hose, and inspect the brushes and tracks for wear. Place the robot back in the pool and ensure it is positioned correctly for its next cycle.

ROBOT PLACEMENT

Always place the robot in the deep end of the pool, upside down initially, then flip it over once submerged to release trapped air. This prevents the robot from floating and ensures it makes contact with the floor immediately. If a robot is consistently missing areas of the pool, check the drive tracks and brushes for wear, and verify the filter is not clogged (which reduces suction and wall-climbing ability).

7.5 Salt Chlorinator Cell Maintenance

Salt cells require periodic inspection and cleaning. Calcium scale builds up on the cell plates over time, reducing chlorine production efficiency. Most modern chlorinators have a self-cleaning (reverse polarity) function, but this does not eliminate all scale buildup.

To clean a salt cell: remove it from the housing, inspect the plates visually. If white scale is visible, soak the cell in a solution of 1 part Hydrochloric Acid to 10 parts water for 5–10 minutes. Do not leave it longer than necessary — excessive acid exposure damages the cell coating. Rinse thoroughly with fresh water before reinstalling. Never scrape or physically remove scale from the plates, as this damages the precious metal coating.

CELL CLEANING SAFETY

Always perform cell cleaning outdoors or in a well-ventilated area. The acid solution will produce fumes and bubbles as it dissolves the calcium. Wear gloves and goggles. Dispose of the used acid solution responsibly — never pour it into stormwater drains.

08 Pump Runtimes & Circulation

8.1 The Single Continuous Block Rule

MANDATORY STANDARD — NO EXCEPTIONS

Pump runtimes must ALWAYS be set as a **single continuous block**. Never split runtimes (e.g., 4 hours morning + 4 hours evening). This is a firm Palatial Pools standard.

Why? Splitting the runtime creates a gap during peak UV and heat hours (typically midday to early afternoon) when chlorine demand is highest. During this gap, the pump is off, water is stagnant, chlorine is being destroyed by UV, and bacteria/algae have an uncontested window to multiply. A single continuous block ensures the water is circulating and the chlorinator is producing chlorine during the most critical hours of the day.

8.2 Recommended Runtimes

SEASON	RECOMMENDED RUNTIME	TYPICAL START TIME	NOTES
Summer (Dec–Feb)	8–10 hours	8:00 AM – 9:00 AM	Peak demand; longest runtime needed
Autumn (Mar–May)	6–8 hours	9:00 AM – 10:00 AM	Reduce as temperatures drop
Winter (Jun–Aug)	4–6 hours	10:00 AM – 11:00 AM	Minimum to maintain circulation
Spring (Sep–Nov)	6–8 hours	9:00 AM – 10:00 AM	Increase as temperatures rise

VARIABLE SPEED PUMPS

For pools with variable speed pumps (e.g., Neptune X-Pro NPX1100), the pump can be set to run at a lower speed for longer periods, which is more energy-efficient. However, the single continuous block rule still applies. A common setup is to run at low speed (e.g., 1,500 RPM) for 10–12 hours rather than high speed for 6 hours. Ensure the flow rate at the chosen speed is sufficient for the chlorinator to operate (most chlorinators require a minimum flow rate).

8.3 Turnover Rate

The pump should ideally turn over the entire pool volume at least once per day, and ideally 1.5–2 times in summer. For a 40,000L pool with a pump flowing at 250 L/min, one full turnover takes approximately 2.7 hours. Running for 8 hours provides approximately 3 full turnovers, which is excellent.

Turnover Calculation Guide

POOL VOLUME	FLOW RATE	TIME PER TURNOVER	TURNOVERS IN 8 HRS	TURNOVERS IN 10 HRS
10,000 L	150 L/min	1.1 hours	7.3	9.1
20,000 L	200 L/min	1.7 hours	4.7	5.9
30,000 L	250 L/min	2.0 hours	4.0	5.0
40,000 L	250 L/min	2.7 hours	3.0	3.7
45,000 L	250 L/min	3.0 hours	2.7	3.3
60,000 L	300 L/min	3.3 hours	2.4	3.0

CALCULATING TURNOVER

Formula: Pool Volume (L) ÷ Flow Rate (L/min) ÷ 60 = Hours per Turnover.
Measure flow rate using an inline flow meter or the pump's performance curve at the current speed setting. For variable speed pumps, note that flow rate changes significantly with RPM — always verify the actual flow at the operating speed.

8.4 Circulation Patterns & Dead Spots

Proper circulation means every part of the pool receives filtered, sanitised water. Dead spots — areas where water stagnates — are breeding grounds for algae and bacteria. Common dead spots include behind ladders, in corners, behind steps, and at the far end from the return jets.

To improve circulation: ensure return jets (eyeball fittings) are angled to create a circular flow pattern around the pool, pointing slightly downward at approximately 45 degrees. In rectangular pools, aim jets to push water in the same rotational direction. If a pool has persistent algae in one area, check whether that area receives adequate water flow.

SUCTION CLEANERS & CIRCULATION

When a suction-side cleaner is connected, it draws water through the cleaner hose instead of the skimmer, which can reduce surface skimming. Some pools have a dedicated cleaner line with a valve to balance flow between the skimmer and cleaner. Ensure this valve is set correctly — the skimmer should always maintain some flow to remove surface debris and prevent the pump from losing prime.

09 Troubleshooting & Decision Trees

9.1 Green Pool

POOL IS GREEN — What do I do?

STEP 1	**Test water.** Check Free Chlorine and pH immediately.
IF	pH > 7.6 → Add HCl Acid to bring pH to 7.2–7.4 FIRST. Chlorine will not work at high pH.
STEP 2	**Shock dose.** Add Liquid Chlorine or Granular Chlorine to achieve Free Cl > 10 ppm (super-chlorination).
STEP 3	**Remove phosphates.** Add Phosphate Remover if phosphates > 0.4 ppm.
STEP 4	**Brush and scoop.** Agitate all surfaces to expose algae to chlorine. Remove as much debris as possible.
STEP 5	**Run pump continuously** for 24–48 hours. Backwash filter every 6–8 hours as it loads up.
STEP 6	**Re-test after 24 hours.** If still green, repeat shock dose. If clearing, add Clarifier to speed up.
IF	CYA > 50 ppm → Chlorine may be locked. Partial drain and refill may be required before treatment will be effective.
IF	Chlorinator is non-functional → Do not waste chemicals on a full restoration. Stabilise water temporarily and recommend equipment repair/replacement first.

9.2 Cloudy Water

WATER IS CLOUDY / MILKY — What do I do?

CHECK	**Filter pressure.** Is it high? If yes, backwash or clean cartridge first.
CHECK	**Free Chlorine.** Is it low? If yes, the cloudiness may be early-stage algae. Shock treat.
CHECK	**pH and TA.** High pH or high TA can cause calcium to precipitate, creating cloudiness.
CHECK	**Calcium Hardness.** If very high (>400 ppm), calcium is likely precipitating. Partial drain and refill.
ACTION	If chemistry is balanced but water is still cloudy, add **Pool Pro Clarifier** to bind fine particles for the filter to catch.
ACTION	For severe cloudiness, use **Pool Pro Flocculant** — add with pump off, let settle 12–24 hours, vacuum to waste.

9.3 Pump Not Priming

PUMP WON'T PRIME — What do I do?

CHECK	**Water level.** Is it below the skimmer? If yes, fill the pool first.
CHECK	**Pump lid o-ring.** Is it cracked, dry, or not seated properly? Lubricate or replace.
CHECK	**Pump basket housing.** Is it full of water? If empty, fill with water from a hose before starting.
CHECK	**Suction lines.** Are all valves open? Is the cleaner hose connected and submerged?
CHECK	**Air bleed valve** on the filter. Open it until water sprays out, then close.
IF	Still not priming → Check for suction-side air leaks (cracked pipe, loose fitting, failed valve seal). May require pressure testing.
IF	Pump hums but impeller doesn't spin → Impeller may be jammed with debris or the motor capacitor may have failed. Turn off and inspect.

9.4 Suspected Leak

SUSPECTED LEAK — What do I do?

STEP 1	**Confirm water loss is abnormal.** Normal evaporation in Melbourne summer is 3–7mm per day. Mark the water level and check after 24 hours with the pump off.
STEP 2	**Bucket test.** Place a bucket of pool water on the step. If pool level drops more than the bucket, it is a leak (not evaporation).
STEP 3	**Pump on vs pump off test.** Mark level with pump on for 24 hours, then pump off for 24 hours. If loss is worse with pump on, leak is likely on the pressure side (return lines). If worse with pump off, leak is on the suction side or in the shell.
STEP 4	**Visual inspection.** Check all visible plumbing, equipment connections, and around the pool shell for wet spots or erosion.
STEP 5	**Refer for specialist leak detection.** Palatial Pools offers acoustic and infrared leak detection from $345. Do not attempt to dig or excavate without proper detection first.

9.5 High Combined Chlorine (Chloramines)

COMBINED CHLORINE > 0.4 ppm — What do I do?

CAUSE Chloramines form when free chlorine reacts with nitrogen compounds (sweat, urine, organic matter). They cause the "chlorine smell" and eye/skin irritation.

ACTION **Breakpoint chlorination.** Shock the pool to raise Free Chlorine to 10x the Combined Chlorine reading. This oxidises the chloramines.

EXAMPLE Combined Cl = 0.89 ppm → Raise Free Cl to at least 8.9 ppm to reach breakpoint.

FOLLOW UP Run pump continuously until Free Cl drops back to normal range. Re-test after 24 hours.

10 Real Job Case Studies

The following case studies are drawn from actual Palatial Pools service jobs. They demonstrate how our procedures, diagnostic approach, and chemical dosing are applied in real-world situations. Study these examples to understand the thought process behind each decision.

Case Study 1: Multi-Visit Restoration — Neglected Quartzon Pool

JOB DETAILS — INV-1302

Pool: Quartzon surface | 40,000L | Salt chlorinator (4,000 ppm target)
Location: Balwyn, VIC
Issue: Pool neglected during property vacancy. Chlorinator clock reset (power off). Thick debris layer on floor. Water chemistry severely out of balance.

Initial Water Test Results

PARAMETER	TARGET	ACTUAL	STATUS
Free Chlorine	~4.0 ppm	0.2 ppm	Critical Low
pH	7.2–7.4	8.4	Critical High
Total Alkalinity	80–120	98 ppm	OK
Cyanuric Acid	30–50	24 ppm	Low
Calcium Hardness	150–200	300 ppm	OK
Salt	3,000–5,000	3,324 ppm	OK
LSI	-0.3 to +0.3	0.72	Scale-forming

Actions Taken

Visit 1: Added 1.3L Pool Pro HCl Acid (pH 8.4 → target 7.2), 500g Pool Pro Stabilised Chlorine (FC 0.2 → shock), 200g Pool Pro Rapid Stabiliser (CYA 24 → target 30+). Extensive scooping, brushing, and vacuuming. Reset chlorinator clock and verified cell operation.

Subsequent Visits: Continued staged debris removal. Applied Clarifier (Diamond tablets) to bind fine particles. Added Trichlor tablets for sustained chlorine. Could not backwash due to drainage issue (water floods under stairs) — all debris removed manually. Documented drainage issue for client to address.

Pump Seizure (Visit 4): Tenant attempted to clean pump basket, caused air ingress. Pump seized. Technician re-primed system, educated tenant on correct procedure.

Key Lessons

Always check the chlorinator clock on first visit to a previously vacant property. When backwashing is not possible, factor in significantly more labour time for manual debris removal. Educate tenants on basic pump basket maintenance to prevent callbacks.

Case Study 2: Green Pool Restoration — Failed Chlorinator

JOB DETAILS — INV-1296

Pool: Tile surface | 40,000L | Salt chlorinator (3,500 ppm target)
Location: Hawthorn, VIC
Issue: Severely neglected green pool. Chlorinator non-functional (salt cell failed). Equipment area in disrepair. Multiple visits required.

Initial Water Test Results

PARAMETER	TARGET	ACTUAL	STATUS
Free Chlorine	~4.0 ppm	0.07 ppm	**Critical Low**
pH	7.2–7.4	8.3	**Critical High**
Total Alkalinity	80–120	144 ppm	**Slightly High**
Cyanuric Acid	30–50	5 ppm	**Critical Low**
Phosphates	<0.4	1.26 ppm	**Very High**
Salt	3,000–4,000	6,587 ppm	**Very High**

Actions Taken

Visit 1: Shock treatment — 500g Stabilised Chlorine, 1.2L HCl Acid, 1kg Rapid Stabiliser, 800mL Phosphate Remover. Extensive debris removal.

Visit 2: Pool responding. FC now 18.2 ppm (very high from shock). TA crashed to 0 — added 4kg pH Buffer. Continued vacuuming.

Visit 3: Continued vacuuming. Vacuum repeatedly clogging due to debris volume. Solid objects (rocks) found in pool.

Visit 4: Salt dropped to 1,038 ppm. CYA spiked to 138 ppm. LSI critically low at -3.0. Pool stabilised but full restoration postponed.

Key Decision

The technician made the correct call to postpone full restoration. With a non-functional chlorinator and CYA at 138 ppm (chlorine lock), continuing to pour chemicals into the pool would waste money and not achieve

a lasting result. The recommendation was to repair/replace the chlorinator and partially drain the pool to reduce CYA before completing the restoration. This is the pragmatic, client-first approach that defines Palatial Pools.

Case Study 3: Diagnostic Service — No Chlorinator, Inoperable Filter

JOB DETAILS — INV-1300

Pool: Vinyl surface | 45,000L | Liquid chlorine (no chlorinator)
Location: Balwyn North, VIC
Issue: Filter inoperable, no chlorinator present, system in poor overall condition.

Water Test Results

PARAMETER	TARGET	ACTUAL	STATUS
Free Chlorine	~4.0 ppm	3.79 ppm	**OK (manually dosed)**
pH	7.2–7.4	8.0	**High**
Combined Chlorine	<0.4	0.89 ppm	**High**
Calcium Hardness	125+ (vinyl)	65 ppm	**Critical Low**
Cyanuric Acid	30–50	5 ppm	**Critical Low**

Actions & Recommendations

The technician completed a digital water test and manually dosed chemicals to stabilise the water under the current system limitations. The key recommendation was that the system requires a **complete equipment upgrade**: new filter, chlorinator installation, and replumb of the equipment area. The existing equipment was assessed as beyond reliable repair. This honest assessment — rather than attempting repeated band-aid fixes — saves the client money in the long run.

Case Study 4: Regular Service with Heat Pump Issue

JOB DETAILS — INV-1301

Pool: Fibreglass surface | 10,000L | Salt chlorinator (3,500 ppm target)
Location: Oakleigh South, VIC
Issue: Heat pump cutting out before reaching set temperature. Low alkalinity. Blocked return jet.

Actions Taken

Standard service completed: vacuum, scoop, brush, cartridge filter cleaned (flow rate confirmed at 205 L/min). Chlorine production increased from ORP 400 to 500. Added 3kg pH Buffer for low TA (72 ppm). Acid drum topped up.

Heat pump: Switched from direct to automatic mode, opened inlet valve further, increased set temperature to monitor. Unit operated normally for remainder of visit. If issue recurs, likely thermal cycling — may need relocation.

Blocked jet: Inspected but could not remove — fitting appears epoxied into fibreglass shell. Removal would risk damage. Recommended specialist camera inspection.

11 Safety & Chemical Handling

11.1 Personal Protective Equipment (PPE)

CHEMICAL	MINIMUM PPE REQUIRED	ADDITIONAL PRECAUTIONS
Hydrochloric Acid	Chemical-resistant gloves, safety goggles, enclosed footwear	Respiratory protection in enclosed spaces. Work upwind. Never pour near face.
Liquid Chlorine	Chemical-resistant gloves, safety goggles	Avoid splashing. Rinse any spills immediately. Stains clothing.
Granular Chlorine	Chemical-resistant gloves, safety goggles	Avoid inhaling dust. Pre-dissolve in bucket of water before adding to pool.
pH Buffer / Soda Ash	Gloves, safety glasses	Dusty — avoid inhaling. Pre-dissolve in bucket.
Phosphate Remover	Gloves, safety glasses	Mild irritant. Wash hands after handling.
Algaecide	Gloves, safety glasses	Follow label instructions. Some formulations foam if overdosed.

11.2 Critical Safety Rules

THE FIVE COMMANDMENTS OF CHEMICAL SAFETY

1. NEVER mix different pool chemicals together. Mixing chlorine and acid produces toxic chlorine gas. Mixing different chlorine types can cause fire or explosion.

2. ALWAYS add acid to water, never water to acid. Adding water to concentrated acid causes a violent exothermic reaction that can splash acid onto you.

3. ALWAYS wear PPE. Gloves and goggles are the absolute minimum for every chemical handling task. No exceptions.

4. ALWAYS store chemicals securely. Keep oxidisers (chlorine) separated from acids. Store in a cool, dry, well-ventilated area. Secure in the vehicle during transport.

5. ALWAYS have SDS available. Safety Data Sheets for every chemical you carry must be accessible in the vehicle or on your phone. If emergency services ask, you must be able to provide them.

11.3 First Aid

EXPOSURE TYPE	IMMEDIATE ACTION	FOLLOW UP
Skin contact (acid/ chlorine)	Remove contaminated clothing. Flush skin with copious water for at least 15 minutes.	Seek medical advice if irritation persists or burns are visible.
Eye contact	Flush eyes with clean water for at least 15 minutes. Hold eyelids open.	Seek immediate medical attention. Do not rub eyes.
Inhalation (acid fumes/chlorine gas)	Move to fresh air immediately. Loosen tight clothing.	If breathing difficulty persists, call 000. Seek medical attention.
Ingestion	Do NOT induce vomiting. Rinse mouth with water. Drink small sips of water.	Call Poisons Information Centre: 13 11 26. Seek immediate medical attention.

EMERGENCY CONTACTS

Emergency Services: 000
Poisons Information Centre: 13 11 26
WorkSafe Victoria: 1800 136 089
Palatial Pools Office: 0452 157 694

12 Quick Reference Cards

These cards summarise the most frequently needed information. Print them separately and keep them on your clipboard for fast access in the field.

QUICK REF — WATER CHEMISTRY TARGETS

PARAMETER	TARGET	ACTION IF LOW	ACTION IF HIGH
Free Chlorine	~4 ppm	Add Stabilised Chlorine or Liquid Chlorine	Let burn off naturally; check CYA
pH	7.2–7.4	Add Soda Ash (pH Increaser)	Add HCl Acid
Total Alkalinity	80–120 ppm	Add pH Buffer (Sodium Bicarb)	Add HCl Acid (lowers both pH & TA)
Calcium Hardness	150–200 ppm	Add Hardness Raiser	Partial drain & refill
CYA	30–50 ppm	Add Rapid Stabiliser	Partial drain & refill (only way to reduce)
Salt	3,000–5,000 ppm	Add pool salt	Partial drain & refill
Phosphates	<0.4 ppm	No action needed	Add Phosphate Remover

QUICK REF — CHEMICAL DOSING (per 10,000L)

PRODUCT	DOSE PER 10,000L	METHOD
Pool Pro HCl Acid	300–350 mL	Pour into return jet area, pump running
Pool Pro pH Buffer	750g – 1kg	Pre-dissolve, pour around edge
Pool Pro Stabilised Chlorine	125g (shock)	Pre-dissolve, pour around edge
Pool Pro Liquid Chlorine	500 mL	Pour directly, pump running
Pool Pro Rapid Stabiliser	50–250g	Pre-dissolve in warm water, add to skimmer
Pool Pro Phosphate Remover	200 mL	Pour near return jet, pump running

QUICK REF — SERVICE VISIT CHECKLIST

#	TASK	DONE
1	Visual inspection on arrival	☐
2	Digital photometer water test	☐
3	Empty skimmer basket(s) & deck canisters	☐
4	Empty pump basket (pump OFF first)	☐
5	Backwash filter / clean cartridge	☐
6	Vacuum, scoop, brush	☐
7	Inspect equipment area	☐
8	Dose chemicals (correct order)	☐
9	Adjust system parameters	☐
10	Check water level	☐
11	Clean/replace robot	☐
12	Final leak inspection	☐
13	Document everything in PoolTrackr	☐

QUICK REF — PUMP RUNTIMES (Single Continuous Block)

SEASON	HOURS	START TIME
Summer	8–10 hrs	8:00–9:00 AM
Autumn	6–8 hrs	9:00–10:00 AM
Winter	4–6 hrs	10:00–11:00 AM
Spring	6–8 hrs	9:00–10:00 AM

NEVER split runtimes. Always a single continuous block covering peak UV hours.

QUICK REF — TROUBLESHOOTING AT A GLANCE

SYMPTOM	MOST LIKELY CAUSE	FIRST ACTION
Green water	Low chlorine + high pH	Lower pH, then shock dose chlorine
Cloudy / milky water	Poor filtration or high pH/TA	Backwash filter, check chemistry
Strong chlorine smell	High combined chlorine (chloramines)	Breakpoint chlorination (shock)
Pump not priming	Air leak, low water, bad o-ring	Check lid o-ring, fill basket housing
Low chlorine despite dosing	High pH, high CYA, or phosphates	Test pH and CYA first
Rapid water loss	Leak (not evaporation)	Bucket test, then refer for detection
Scaling on surfaces	High calcium, high pH, positive LSI	Lower pH, check calcium hardness
Staining (brown/green)	Metals (iron/copper) in water	Test metals, use sequestrant

QUICK REF — pH vs CHLORINE EFFECTIVENESS

PH	ACTIVE CHLORINE %	VERDICT
7.0	73%	Excellent (slightly acidic)
7.2	63%	Ideal
7.4	50%	Good
7.6	33%	Declining
7.8	22%	Poor
8.0+	**<15%**	**Ineffective — lower pH first!**

A Appendix — Compliance & Standards

A.1 Relevant Australian Standards

STANDARD	TITLE	RELEVANCE
AS 3633	Private Swimming Pools — Water Quality	Defines water chemistry parameters for residential pools. Our target ranges are based on this standard.
AS 1926	Swimming Pool Safety	Pool fencing and barrier requirements. While not our primary role, report obvious non-compliance to the client.
AS/NZS 3500	Plumbing and Drainage	Applies to all plumbing works on pool equipment. Licensed plumber required for regulated works.

A.2 SPASA Victoria

SPASA (Swimming Pool and Spa Association) Victoria is the peak industry body for the pool and spa industry in Victoria. Palatial Pools references SPASA guidelines for water balance, equipment standards, and best practice procedures. For the latest fact sheets and technical resources, visit **www.spasa.com.au**.

A.3 WorkSafe Victoria

All chemical handling must comply with WorkSafe Victoria regulations. This includes maintaining Safety Data Sheets (SDS) for all chemicals carried in the service vehicle, using appropriate PPE, and following safe storage and transport procedures. For chemical safety alerts and guidance, visit **www.worksafe.vic.gov.au**.

> **CHEMICAL TRANSPORT IN SERVICE VEHICLES**
>
> All chemicals must be transported in a secure, upright position in a ventilated compartment. Acids and oxidisers (chlorine) must be stored in separate, sealed containers and never transported together in the same tub or tray. Ensure lids are tight and containers are not leaking before driving. Keep a spill kit in the vehicle at all times. In the event of a spill, contain it immediately, ventilate the area, and do not allow chemicals to enter stormwater drains.

A.3.1 Record Keeping Requirements

Under Victorian regulations and SPASA best practice, service records must be maintained for all pool maintenance activities. This includes water test results, chemical additions (product name, quantity, date, time), equipment inspections, and any issues identified. Palatial Pools uses PoolTrackr as the primary record-keeping system. All records must be entered before leaving the job site. These records may be required in the event of a health incident, insurance claim, or regulatory inquiry.

A.3.2 Seasonal Maintenance Calendar

MONTH	KEY ACTIONS
September	Spring startup: full water test, increase pump runtime, check chlorinator cell, clean filter, inspect equipment
October–November	Increase chlorinator output, verify CYA levels before summer UV, check salt levels
December–February	Peak season: maximum runtimes, weekly monitoring, watch for phosphate spikes from garden runoff
March–April	Reduce runtimes gradually, full water test, address any end-of-summer chemistry drift
May–June	Winterise: reduce pump runtime to minimum, lower chlorinator output, clean and inspect equipment
July–August	Minimal maintenance: monthly check, ensure pump is running, prevent stagnation

A.4 Palatial Pools Business Details

Business Name	Palatial Pool Services
ABN	41 690 936 095
Location	Abbotsford, Melbourne, VIC 3067
Phone	0452 157 694
Email	service@palatialpools.com.au
Website	www.palatialpools.com.au
Hours	Mon–Fri 9am–6pm | Sat 9am–5pm
Guarantee	10 Year Workmanship Guarantee

This manual is the property of Palatial Pool Services. Version 1.0 — April 2026.
Information sourced from SPASA Victoria, WorkSafe Victoria, Pool Pro Australia, and internal Palatial Pools operational data.
For updates or corrections, contact service@palatialpools.com.au.